

administration or clustered support for high availability.

In this article, we will install GlassFish-Installer-v2.1.1 Build 31 on Red Hat Enterprise Linux 6.1 (32-bit).

## Pre-requisites

First, set the IP address by clicking on *System -> Preferences -> Network Connection*. Next, configure the host file with *Configuration File - /etc/hosts*. Now, turn off the firewall by clicking on *System -> Administration -> Firewall*. Then, enforce SELinux as follows: *Command - getenforce; Configuration File - /etc/sysconfig/selinux*. Finally, install Java. In our case, we will be installing *jdk-1\_5\_0\_20-linux-i586.rpm*.

## Java installation

Given below are the steps for installing Java.

1. Provide recursive permission to the Java installer as follows:

```
chmod -R 755 jdk-1_5_0_20-linux-i586.rpm
```

2. Install Java with the following command:

```
rpm -ivh --aid --force jdk-1_5_0_20-linux-i586.rpm
```

3. Create symbolic links:

```
ln -s /usr/java/jdk1.5.0_20/bin/java /usr/bin/java
ln -s /usr/java/jdk1.5.0_20/bin/javac /usr/bin/javac
```

4. Verify the Java installation with the following commands:

```
a. java -version
b. which java
c. whereis java
d. java
e. javac
```

## Installing the GlassFish application server

1. Provide recursive permission to the GlassFish installer, as follows:

```
chmod -R 755 glassfish-installer-v2.1.1-b31g-linux.jar
```

2. Create a directory by any name on the '/' file system:

```
mkdir /appinstall
```

3. Browse into the newly created directory and then run the GlassFish Jar installer using the following Java command:

```
cd /appinstall
java -Xmx256m -jar /opt/Setup/glassfish-installer-v2.1.1-b31g-linux.jar
```



**Note:** In our case, the GlassFish installer is kept under '/opt/Setup'

4. Press 'A' to 'Accept the License Agreement' and installation then proceeds. When you see the 'Installation complete' message followed by a return of the root prompt, you know that the installation is done.
5. As observed earlier, a directory named *glassfish* is created under the *Appinstall* directory.

## Building the GlassFish application server

1. Set the executable permission to *lib/ant/bin* modules under *glassfish* directory:

```
chmod -R +x lib/ant/bin/
```

2. Using the 'ant' executable (located under '*appinstall/glassfish/lib/ant/bin*'), run *setup.xml* (located under '*appinstall/glassfish*') to build GlassFish:

```
lib/ant/bin/ant -f setup.xml
```

3. Under the GlassFish directory, there are two *GlassFish setup.xml* files:
  - *setup.xml*—for building a standalone GlassFish environment.
  - *setup-cluster.xml*—for building a clustered GlassFish environment.

You know the build is successful when you see the following message:

```
'BUILD SUCCESSFUL
Total time: XX seconds'
```

...followed by a return of the root prompt.

4. Make a note of the following port numbers, which will be required later:
 

|               |      |
|---------------|------|
| Admin console | 4848 |
| HTTP instance | 8080 |
| JMS           | 7676 |
| IIOP          | 3700 |
| HTTP_SSL      | 8181 |

## Starting and stopping the GlassFish application server

Browse to the *bin* directory under *glassfish*:

```
cd /appinstall/glassfish/bin
```

Run the following command:

```
./asadmin
```